BayesSSD – Bayesian Sample Size Determination for Multilevel Models

Workshop in Utrecht University
Bayes Group
19 March 2026

Utrecht University

NWO

Workshop structure

- Theoretical background
- Exercise 1: Attrition
- Cluster Randomized Trials
- Exercise 2: SSD for a trial

The Problem

Lack of free software for Bayesian sample size determination for multilevel models

High threshold for using Bayesian hypothesis testing

Bayesian vs. frequentist power

- Frequentist power

	H0 is true	H0 is false
H0 not rejected	✓	Type-II error
H0 rejected	Type-I error	✓

Power = 1 – P(type-II error) = 1 – beta

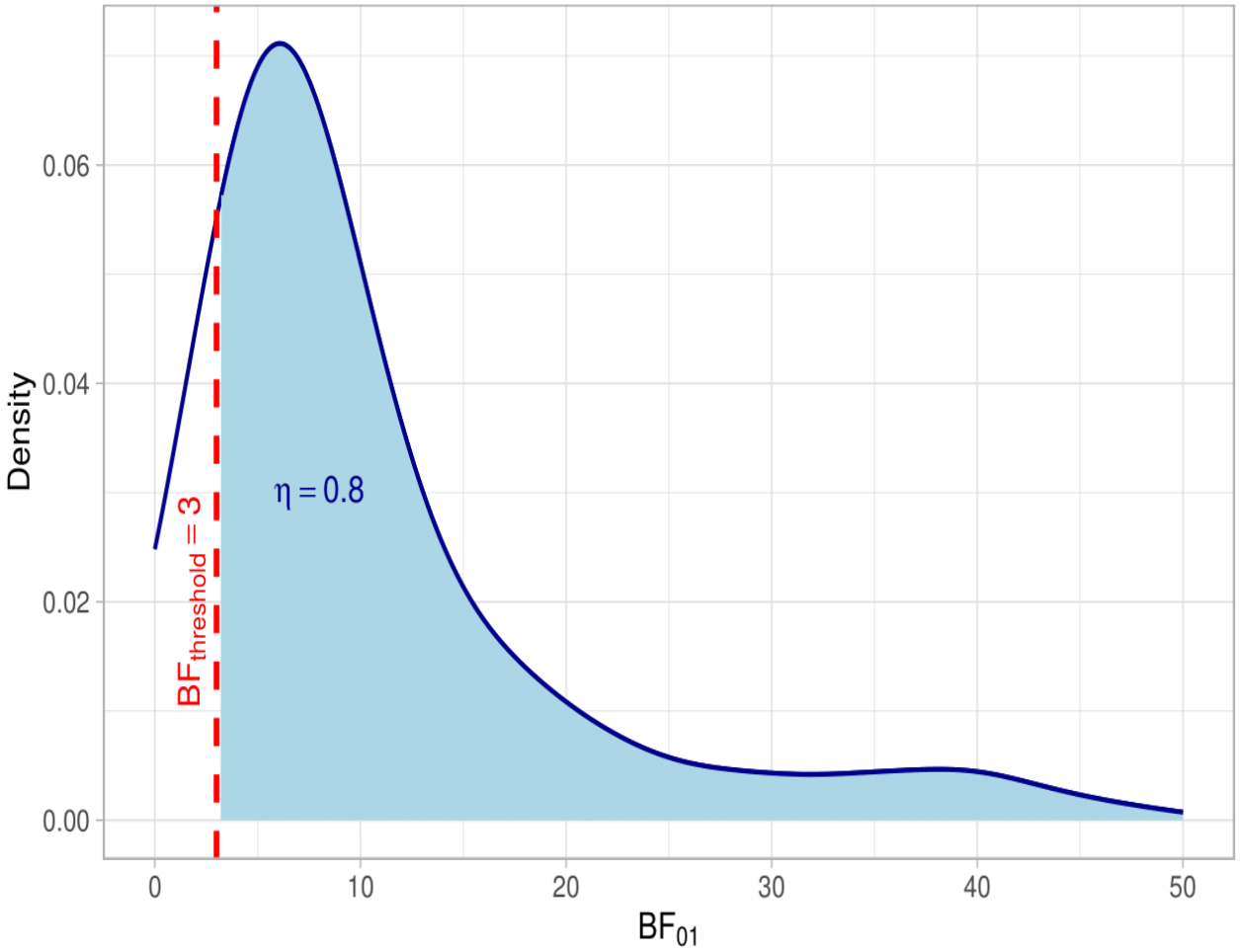
Bayesian vs. frequentist power

- Bayesian power
  - *Assurance:* unconditional probability of a successful intervention averaged over the parameter space of a prior treatment effect and other nuisance parameters. (O'Hagan, 2005; Chen et al., 2018) **computationally intensive**
  - Pragmatic: Power = P(BF1c > threshold | H1)

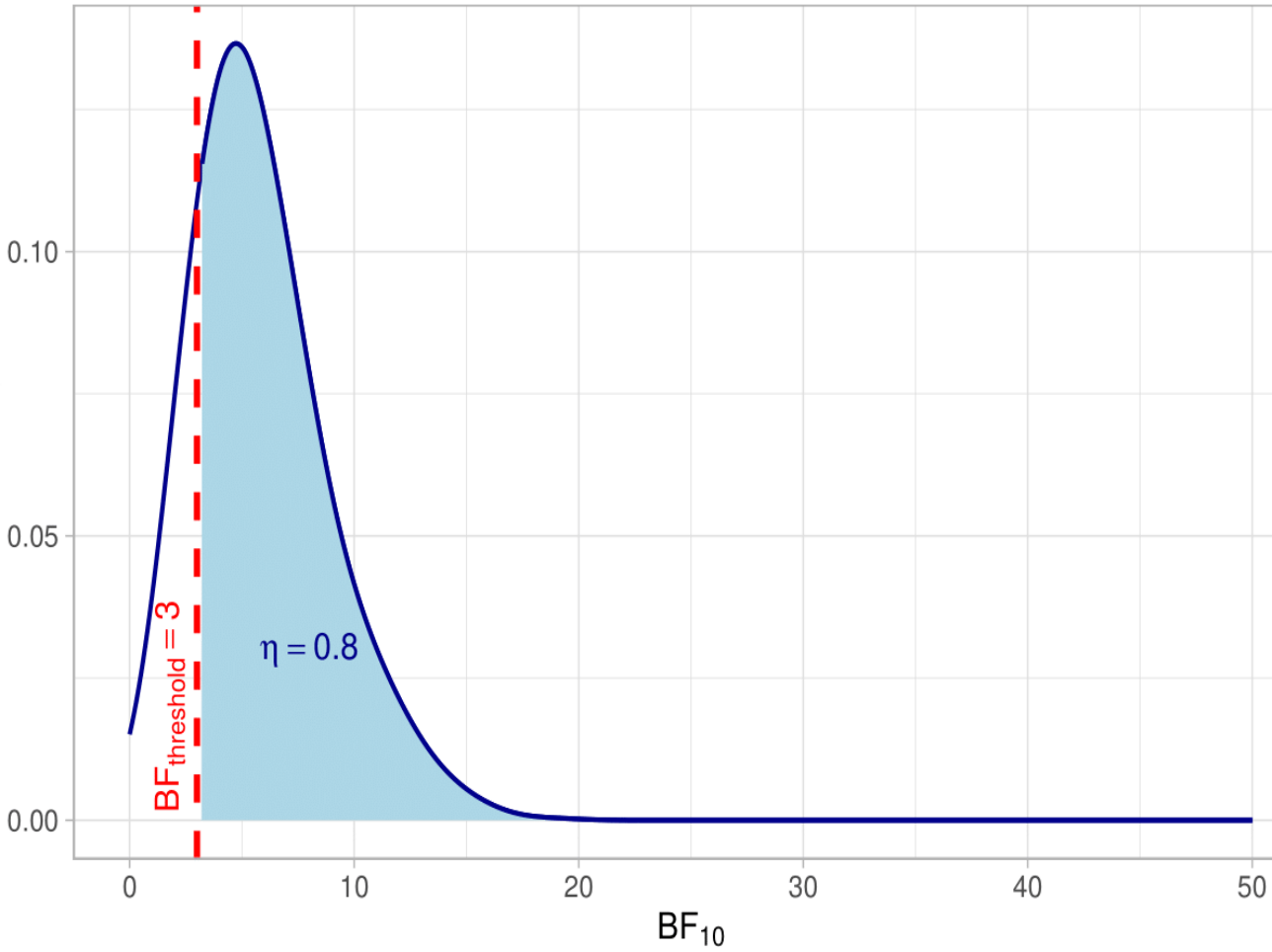
	H1 is true	H1 is false
BF1c > threshold	✓	Type-I error
BF1c < threshold	Type-II error	✓

Bayesian vs. frequentist power

H_0=true



H_1=true



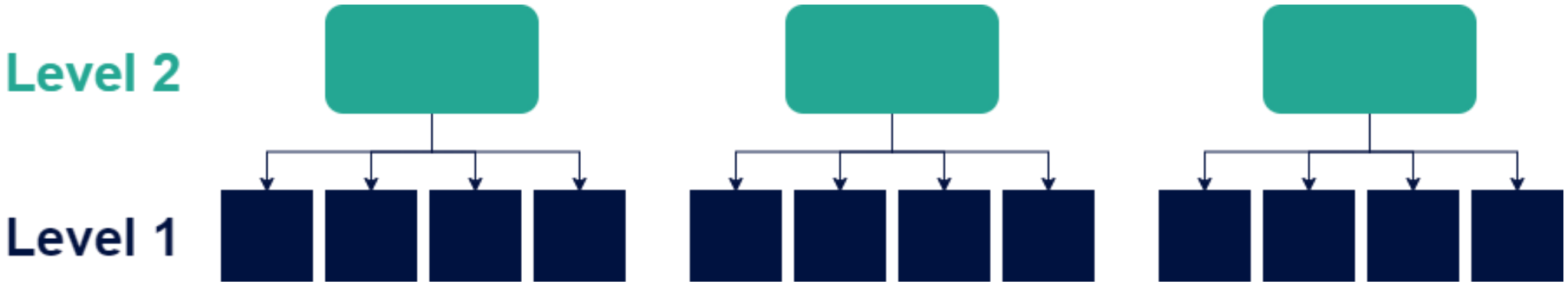
Multilevel Model

- Hierarchical structure
- Correlation of data

$$\beta_{oj} = \gamma_{00} + \gamma_{01} Z_j + u_{oj}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11} Z_j + u_{0j}$$

$$y_{ij} = \beta_{oj} + \beta_{1j} X_{ij} + e_{ij}$$



Approximate Adjusted Fractional BF

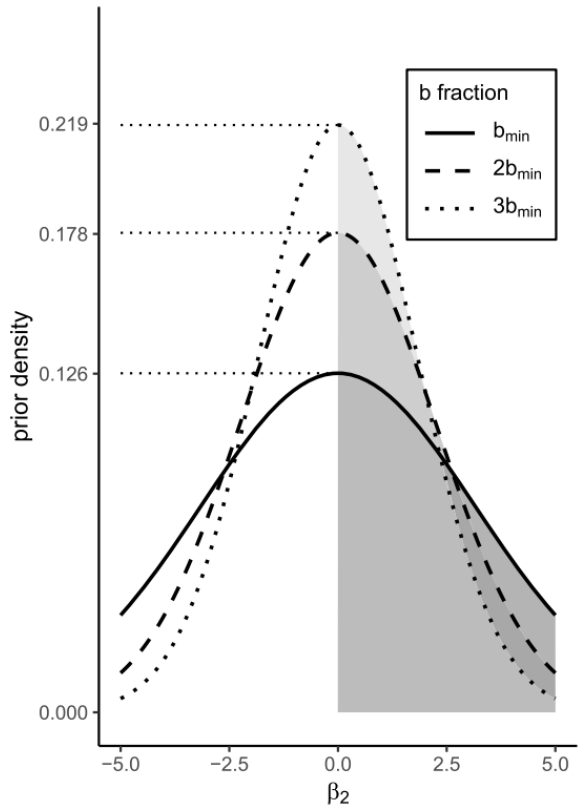
(Gu et al., 2018)

- $BF_{1u} = fit_1 \ / \ complexity_1$
- $BF_{12} = BF_{1u} \ / \ BF_{2u}$

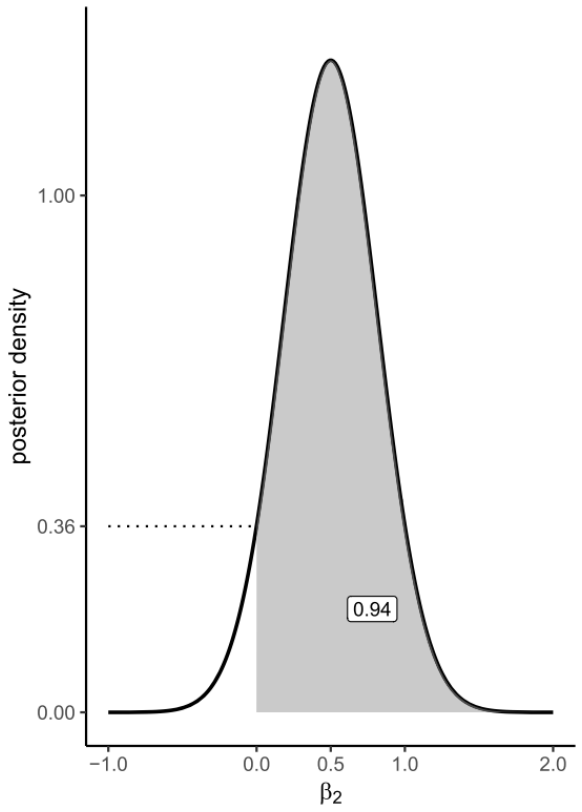
- Default normal prior constructed using fraction of the data (*b*)

b-fraction sensitivity

(see left plot)



(a) *Prior distributions and complexities of $\mathcal{H}_0$ and $\mathcal{H}_1$. The solid, dashed and dotted line corresponds to the prior distribution using b_{min}, $2b_{min}$ and $3b_{min}$, respectively. The horizontal dotted lines indicate the complexity of $\mathcal{H}_0$ under each prior (0.126, 0.178 and 0.219) while the shaded area under the curve indicates the complexity of $\mathcal{H}_1$ under each prior (always 0.5).*

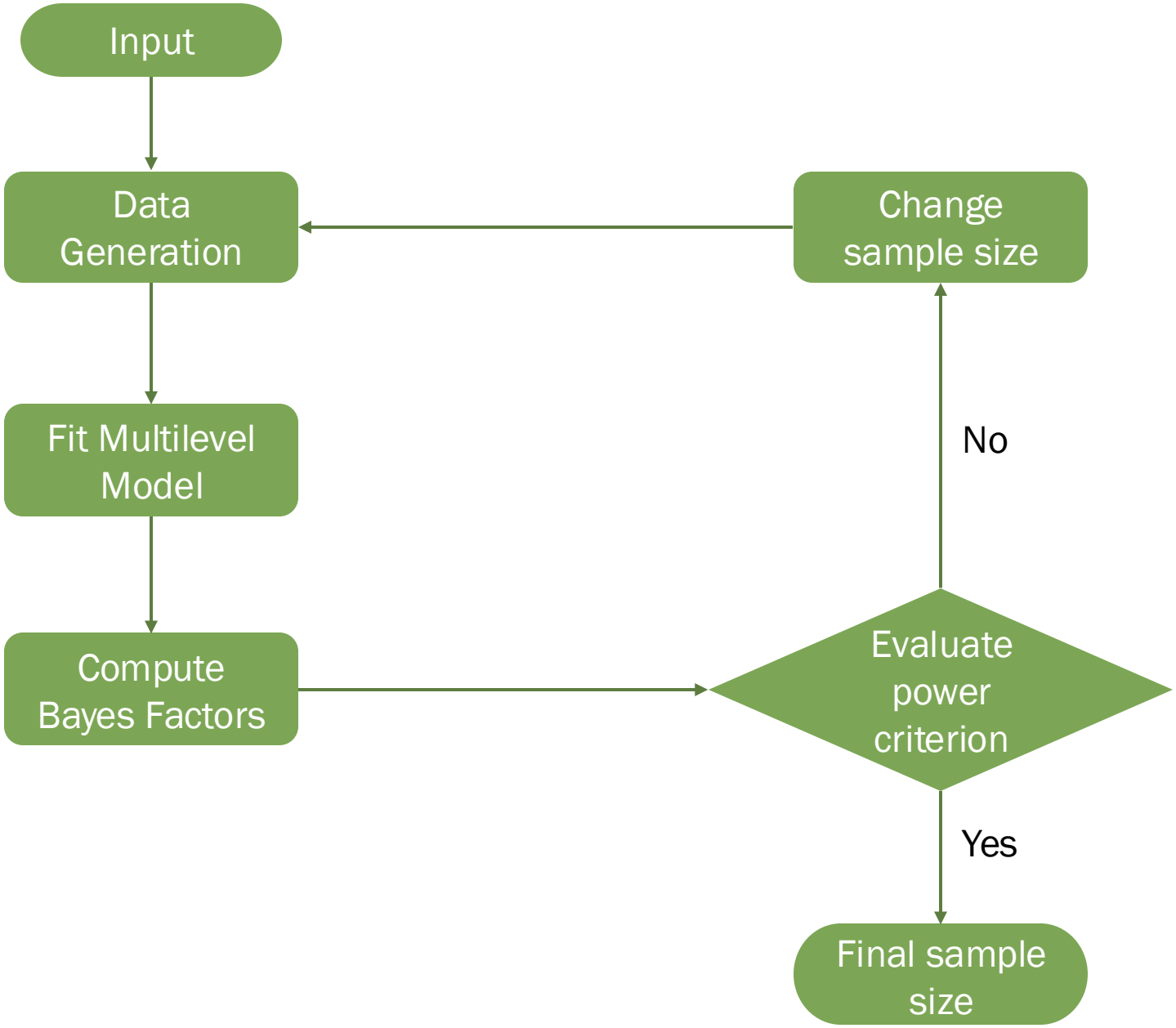


(b) *Posterior distribution for β_2 with $\hat{\beta}_2 = 0.5$ and $\sigma^2_{\hat{\beta}_2} = 0.1$. The fit for $\mathcal{H}_0$ is indicated by the horizontal dotted line and equal to $g_u(\beta_2 = 0 \mid X) = 0.36$. The fit for $\mathcal{H}_1$ is indicated by the shaded area and equal to $g_u(\beta_2 > 0 \mid X) = 0.94$.*

Figure taken from Lösener & Moerbeek, 2025

Simulation Algorithm

- Generative model based on input values
- For candidate N, simulate data, fit model, calculate BF, apply decision rule
- Binary search
- Convergence if …



Binary Search

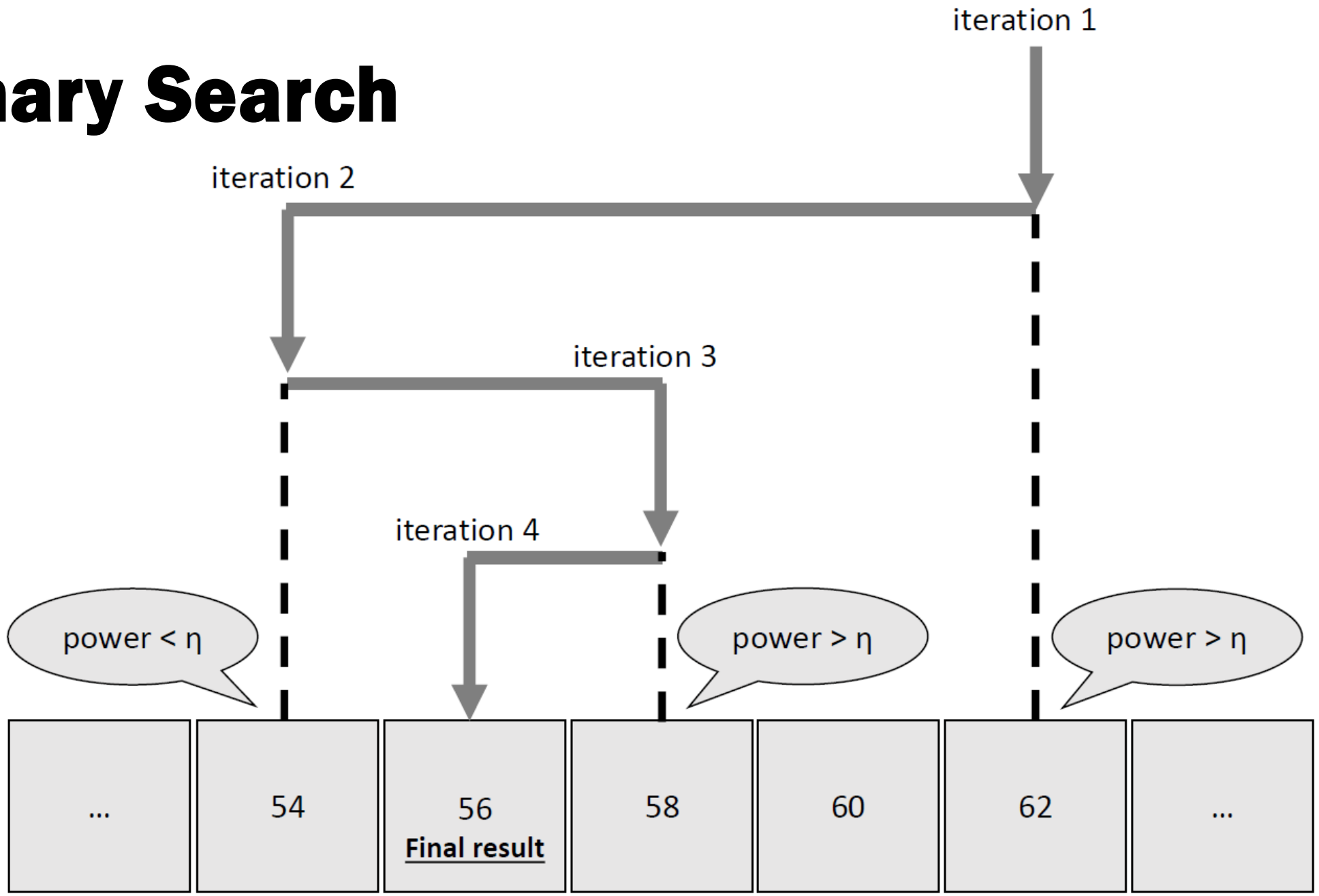
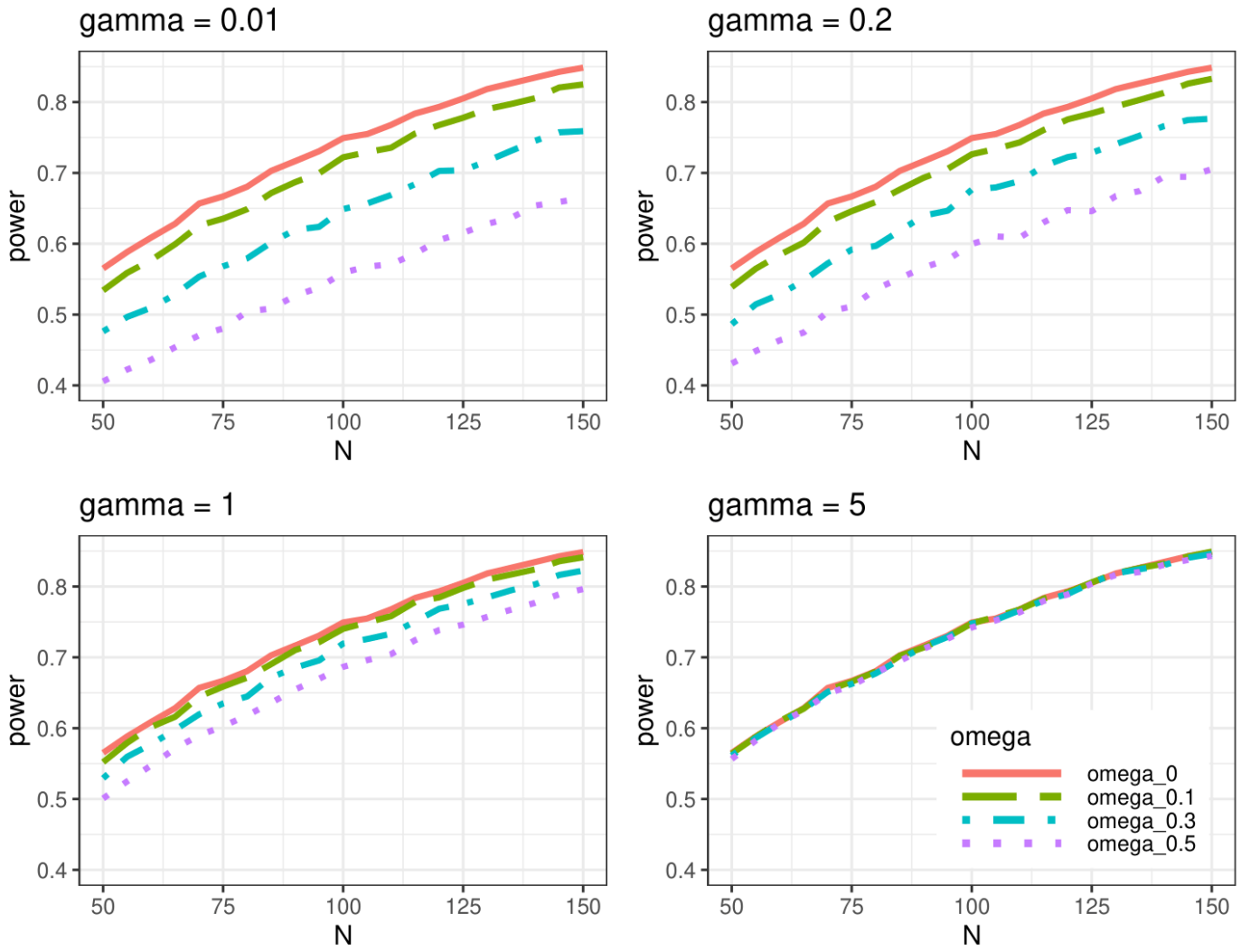


Figure taken from Lösener & Moerbeek, 2025

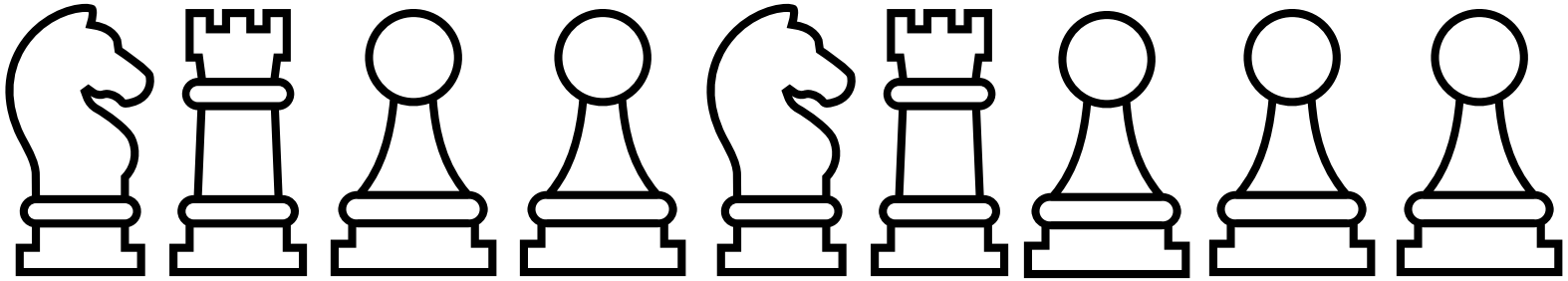
Attrition and Power

- Attrition negatively impacts power (shocking)
- The *timing* of drop-out matters!

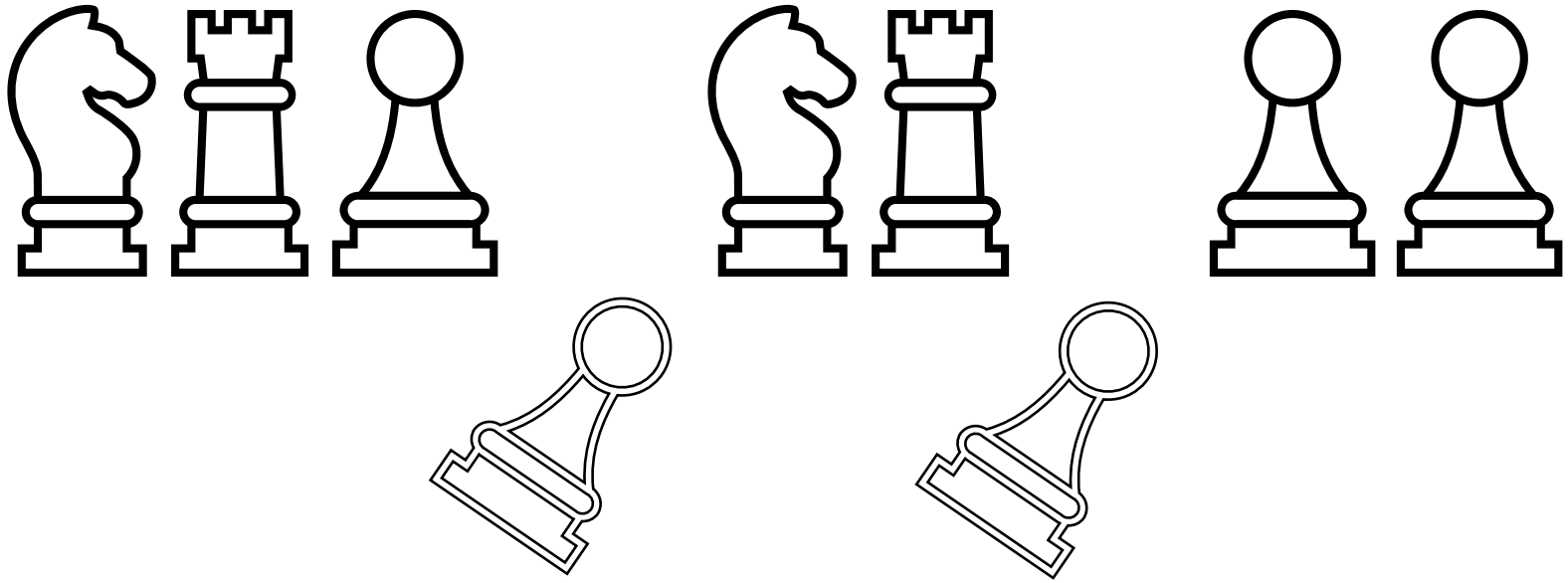
→ The earlier, the worse



Exercise 1 – Model Attrition with Survival Functions



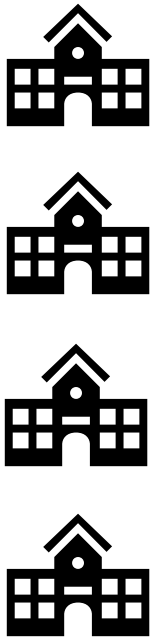
Exercise 1 – Model Attrition with Survival Functions



Cluster Randomised Trials

What is it?

Treatment



Control

Sample Size Determination

- Effect size
- Intraclass correlation coefficient (ICC)
- Bayes factor threshold
- Bayesian power (η)
- Pair of hypotheses
- Fixed cluster size or number of cluster

Simulation Results

- Sample size increases with
  - High ICC
  - Small effect size
  - High Bayes factor threshold
  - Including null hypothesis
- It is easier to reach the power criterion when increasing the number of clusters.
- In some scenarios the power criterion was not met after large cluster size (1000).

Sample Size Determination for Cluster Randomised Trials with Bayes Factor

Results Information

The sample size to be found given the other one is fixed:

- (●) Number of clusters
- ( ) Cluster sizes

Type of hypotheses to compare:

- (●) Null vs. Informative
- ( ) Informative vs. Informative

Cluster sizes: 5

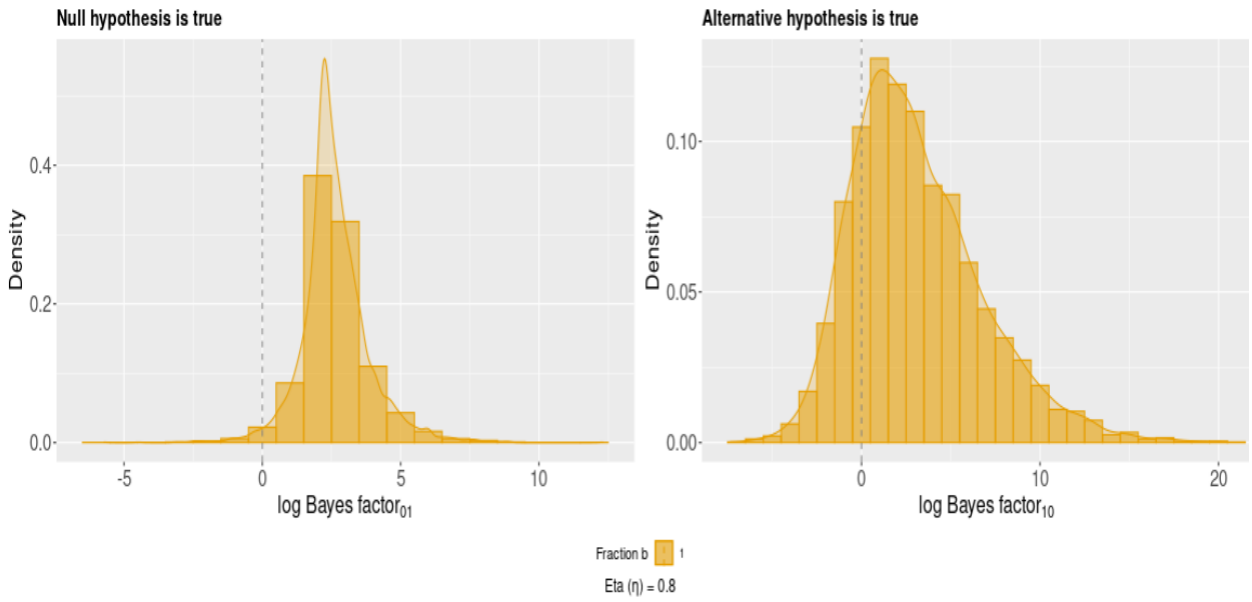
Intraclass correlation: 0.025

Effect size: 0.2

Bayes factor threshold: 1

Fraction b in plots:

- [x] 1
- [ ] 2
- [ ] 3



Final Sample Size

b	Number of clusters	P(BF.01 > thresh)	P(BF.10 > thresh)
1.000	126.000	0.982	0.807
2.000	114.000	0.971	0.801
3.000	108.000	0.963	0.803

```r

get_sample_size <-SSD_crt_null(eff_size = 0.2,
                               n1 = 10, ndatasets = 1000,
                               rho = 0.05,
                               BF_thresh = 3,
                               eta = 0.8,
                               fixed = "n1",
                               b_fract = 3)

```

```
Final sample size
=================
Hypotheses:
    H0: Intervention=Control
    H1: Intervention>Control
**********************************************************************
     b         n1        n2        P(BF.01 > 3 | H0) P(BF.10 > 3 | H1)
[1,] "  1.000" " 10.000" "104.000" "  0.947"         "  0.802"
[2,] "  2.000" " 10.000" "100.000" "  0.919"         "  0.806"
[3,] "  3.000" " 10.000" " 96.000" "  0.879"         "  0.806"
**********************************************************************
n1: Cluster sizes
n2: Number of clusters
```

Take-home Messages

- There now is a package for Bayesian SSD in multilevel scenarios!

→ *BayesSSD*

- Sensitivity analysis is important

- We look forward to your feedback and comments!

References

Chen, D. G., Fraser, M. W., & Cuddeback, G. S. (2018). Assurance in intervention research: A Bayesian perspective on statistical power. *Journal of the Society for Social Work and Research*, *9*(1), 159-173.

Gu, X., Mulder, J., & Hoijtink, H. (2018). Approximated adjusted fractional Bayes factors: A general method for testing informative hypotheses. British Journal of Mathematical and Statistical Psychology, 71(2), 229-261.

Lösener, U., & Moerbeek, M. (2025). Bayesian sample size determination for longitudinal intervention studies with linear and log-linear growth. Behavior Research Methods, 57(9), 239.

O'Hagan, A., Stevens, J. W., & Campbell, M. (2005). Assurance in clinical trial design. *Pharmaceutical Statistics*, *4*, 187–201.